

Tian Qin

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Research Interests

Model predictive control, adaptive/fuzzy PID, autonomous mining trucks, vehicle dynamics and model calibration, data-driven/reinforcement-learning-based control for safety-critical autonomous systems.

Education

China University of Mining & Technology, Beijing (CUMT 211) Sep. 2023 - Sep. 2025
Master: Control Science and Engineering | GPA: 3.59/4.0
Supervisor: Prof. Kehu Yang, Lab of Intelligent Energy System and Autonomous Mining
Coursework: Linear System Theory (96), Adaptive Control (94), Advanced Signal Processing (93)
Sichuan University, Chengdu (SCU 985) Sep. 2015 - Jun. 2019
Bachelor: Electronic and Information Engineering

Publications

Journal

- **Tian Qin**, Desheng Zhu, Chunhui Wang, Peize Yang, Kehu Yang. *Dual-Loop Fuzzy-PID Acceleration Tracking Controller for Autonomous Mining Trucks under Variable Payload Conditions*. Coal Engineering, accepted. First-author work on variable-payload longitudinal control; achieved 0.49 km/h maximum speed error and 0.103 m/s² maximum acceleration error, with speed and acceleration PTP errors improved by 8.77% and 13.30% compared with MPC.

Conference

- **Tian Qin**, Lei Qiu, Jingchuan Chen, Hongyi Fu, Desheng Zhu, Kehu Yang. Double-Layer Following Controller for Autonomous Vehicles. 36th Chinese Control and Decision Conference (CCDC), IEEE, 2024, pp. 908-913. First-author two-layer following controller combining DWA local planning and fuzzy-adaptive PID in ROS/VTD; lateral and longitudinal acceleration RMS improved by 19.5% and 37.3% compared with PID-Stanley.
- Chunhui Wang, Desheng Zhu, **Tian Qin**, et al. Lateral Trajectory Tracking of Autonomous Mining Trucks Using MPC with Adaptive Load Compensation. Chinese Control Conference (CCC), 2025. Contributed mining-truck control context and experiment support for load-adaptive MPC in Simulink/TruckSim; full-load lateral MAE was reduced by 33.33% compared with LQR.
- Peize Yang, Desheng Zhu, **Tian Qin**, et al. Adaptive Heading Tracking Algorithm Based on Vehicle Dynamics Model. Chinese Control Conference (CCC), 2025. Contributed vehicle-dynamics control context and manuscript support for heading-corrected pure pursuit; average lateral error decreased from 0.462 m to 0.237 m in TruckSim/Simulink validation.

Research Experience

Key Technologies and Application Demonstration of Autonomous Transportation and Loading/Unloading System for Large Open-pit Mine Robots

National Key Research and Development Program of China Nov. 2022 - Oct. 2025

- Built adaptive/fuzzy-PID longitudinal-control algorithms for low-speed unmanned mining trucks, focusing on speed and acceleration tracking under variable-payload operating conditions.
- Extended this foundation toward load-adaptive MPC for lateral trajectory tracking, using vehicle dynamics and changing payload parameters to improve robustness in open-pit mining scenarios.
- Framed the next research step around model calibration, parameter identification, and learning-based/data-driven control for safety-critical autonomous systems.

Work Experience

- Mindway Science and Technology Co., Ltd. - Beijing**

Oct. 2025 - Present
- Control Algorithm Engineer*
 - Calibrated vehicle-response parameters from field data for lateral lane tracking, improving controller behavior across mine-road curvature and surface variation.
 - Supported model-parameter tuning and controller validation for longitudinal speed and braking across slopes, payload changes, and mine-road conditions.
 - Built remote safety-control strategies covering fault diagnosis, emergency shutdown, and operational interlocking for autonomous mining-truck operation.

- China Aerospace Science and Industry Corporation (CASIC) - Wuhan**

Jul. 2019 - Aug. 2023
- Hardware Engineer*
 - Participated in mine-site intelligent transformation work for unmanned mining-truck systems, with exposure to large-fleet deployment requirements.
 - Designed PLC controllers and safety-redundancy systems for industrial vehicle-control applications.
 - Supported on-site vehicle testing and troubleshooting, including communication-disconnection and integration issues.

Selected Competitions

- China Intelligent Connected Vehicle Challenge**

Oct. 2023 - Nov. 2023
- Contribution Award (Top 5%)*
 - Developed an adaptive cruise controller in VTD, combining DWA and fuzzy-PID control; lateral and longitudinal acceleration RMS values were kept below 0.1 m/s² and 0.06 m/s².
- Autopilot Challenge of Vehicle-Road-Cloud Integration**

Mar. 2024 - Jun. 2024
- Innovation Award (Top 5%, ranked 7th of 118 teams)*
 - Integrated an LQR controller on a real autonomous-vehicle platform for parking and obstacle-avoidance verification.
- National Post-Graduate Mathematical Contest in Modeling**

Oct. 2024 - Nov. 2024
- Participation Award (Top 20%)*
 - Optimized Steinmetz-equation fitting for magnetic components and improved model accuracy by 5%.

Technical Skills and Languages

- Programming:** C++, Python, MATLAB
- Autonomous driving and simulation:** Apollo, ROS, Linux, VTD, TruckSim, CarSim
- Control and modeling:** MPC, LQR, fuzzy-PID, vehicle dynamics, model calibration, parameter identification
- Languages:** Mandarin (native), English (IELTS 6.5; Reading 8.0)